

# ActInf Livestream #021.01 ~ John Boik

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there we go hello and welcome everyone it is tuesday april 6 2021. and it is active inference livestream uh series that we're going to be kicking off with john boykier this is going to be really interesting because we're going to have three discussions with john and i and interactive jam boards for people who are working with us live or asynchronously and then we're going to be having two group discussions later so it's going to be a really interesting look through a really rich world view and hopefully a paradigm that we can be starting to think about enacting or learning more about what it might look like in reality so let's just get right into this interesting set of topics i'm daniel and i'm a postdoc in california and i'll pass it to john just to introduce himself and this series thanks for coming on john thank you daniel i appreciate it i'm happy to be here with you actually um so we're going to talk in this series about a series of papers that i just published in the in the journal sustainability that that series is titled science driven societal transformation and uh it covers a lot of material all three papers are kind of lengthy so that's one of the reasons that we wanted to kind of have some time together talk out loud and cover some of the some of the main points i am a data science by training just as a little background actually phd and and cancer biology essentially and uh uh also i'm a courtesy faculty at oregon state university so these pape this series of papers is published under uh my affiliation with osu and uh they just came out this this last uh at the end of last year and this this spring so all right so uh i think the jamboard has links i won't bother to give the links to them there's also a little short survey a short summary paper uh in sciencex that the link is there so you can also check that out and uh i would like to say that for for additional information you can go to the project web page which is brunch principled societies project dot org and there you would find links for all the papers and and a bunch of other material also a uh a a live simulation model you could play with too so john let me just ask about the three paper format because like three first author papers in sequence it's kind of a uh there has to be a story there so just the story is the story is i'm verbose no there's there's more to it than that uh you know um i i i had some ideas in mind i mean i i mean obviously that's what the series is about but i had some ideas in mind of of what transformation might look like and how we

might get there but the but i think the ideas are relative are fairly unusual and i i didn't want to just put them in a short paper because without background i think that the ideas could have been easily misinterpreted or misunderstood so i i felt it necessary to explain the background in more detail about what what the proposal actually is and what the concept is it's a it's a the the whole series as a whole is kind of a proposal for an r d project science driven r d project aimed at societal but it's a little bit more than that actually it's it's a it's a it's a way to think about transformation a way to a context for how we might go about uh societal transformation and what that looks like and why we would even want to bother with the concept in the first place so uh so it just was too much to fit in one paper and i i had originally written it and one put it all in one paper and then i could see that wasn't going to work out it was going to be way it's way too long so i split it up and uh i'm very happy that the journal sustainability was willing to publish this they were quite interested so uh so that all worked out really quite well and maybe i should say too this work represents about 10 years of my efforts this is kind of the in a sense the culmination of 10 years of my thinking on this topic cool really interesting and i'm sure we'll hear more about some of those threads so just the first slide on this jamboard which people are welcome to join if they're watching live or if they're looking at the replay the first slide is capturing how we're gonna go through these three parts of the triptych the paper series in order at this same time slot 3 p.m pacific for the next three weeks and then we're going to be having those group discussions so we're going to kind of pour over these papers but who knows maybe take a few tangents as well and this principled societiesproject.org website is where more information can be found so let's just maybe start with a little background before we jump into the paper on just what collaborative dialogue has to do with all of this why are we structuring it this way uh in experimenting with it this way and just how does dialogue fit in in specific or in general to what you're talking about and thinking about well um you know as i mentioned the the series is kind of r d proposal for an r d project and while it's important that that the concept as a whole move forward at least it's important to me and funding you know we secure some funding for this project so it can move forward perhaps the biggest thing that has to happen is just the concept a few concepts a few primary concepts that i talk about just getting those concepts articulated out to the science world and to the public that may you know if i die and that's all the uh that's all that ever happens from this just that those concepts become talked about more more frequently in the public discussion then i probably have done my job obviously i hope it goes a little further than that but that's the place to start and maybe that's the most important thing um where uh would just citizen

science or participatory science dialogue with really uh inclusive participation play a role in the r d programs of the future in what you're kind of thinking about yeah so so um i i i framed this this r d program that is it's conceptual at the time it's not funded yet you know i'm hoping that we can secure funds but i frame it as a partnership between this global science community and local communities so it's very so dialogue with the public and within the science community and among interested stakeholders is extremely important in this um i i i you know to me science has a role in in such a r d program because science is really the you know the where we would turn to answer some really difficult questions if you wanted to build a simulation model of how environmental or environmental or economic uh outcomes might be given you know a b c and d well then you know that's a that's a technical those are technical questions um if you're if you're asking how can we measure how can what kind of metrics are reasonable for environmental and social well-being those are largely scientific questions you know the math can be complicated for example but the questions of you know how do what do we want what do people want how how how do they want their light you know how do they want to live their lives in in society those are questions for you know for the public and for communities especially um i i the the intention of the r d program is not to develop one size fits all solution you know to trial it in a local community and then spread it everywhere that's not at all the idea the idea is that this is an ongoing learning process a true partnership between uh local communities and the science and the science community and there would be just a million sorts you know experiments that one might might might run uh to to improve the kinds of societal systems that we have or that were you know that we're proposing uh develop a new system try it out see how it works gather data you know do another experiment uh all within the partnership of at with local communities at the local community level i think maybe you know i since i know this stuff so well i think that maybe i'm i'm already jumping ahead of some some uh some concepts that might this all might become clear as we talk a little bit more nice yeah well this slide was just to show that there's a bunch of questions that could almost apply to every stage of this model and to every stage of this conversation like what are we doing here what is our goal here what are the next steps and how is active inference coming into play not that active inference good not active inference bad you know we're gonna kind go a little bit deeper than that but what does this have to do with actually having impact will be something that will be cool because what we're talking about is really new syntheses of the theoretical and the applied so that means that we're always like looking to staple across the divide in new ways with the questions that we're asking and the teams that we're working on

the projects and everything so maybe we can go to the fourth slide and to kind of frame this research line what is the issue yeah yeah so so obviously the word transformation is in the title of the series so this you know the general topic is societal transformation and although that term alone is a little bit you know people have different ideas of what societal transformation means so i want to make a few things clear i especially in the second paper i make the distinction between reform and trends and transformation and by reform i mean anything that would you know improve our government system and prove our our economic system improve our legal system you know one example might be uh levying taxes on the wealthy or something and then you know using those fees to to provide medical services or something or altering how long a representative can be in and you know in congress in the legislature or something like that or ways to vote or things like that those really those are all what i would consider reforms and i'm interested in a different question i'm interested in the question out of all conceivable ways to organize the societal systems and by societal systems i really focus on a few of them uh governance systems economic systems uh legal systems educational systems and i think maybe one or two others out of out of that i i view those systems as the cognitive architecture of a society that is that's how society thinks through those systems it it learns it adapts it decides and and evolves kind of through that kind of cognitive architecture and my question is out of all possible conceivable ways out of all con for example out of all conceivable economic systems what ones might be best for uh for demonstratively showing that that they're you know they excel at improving or maintaining um social environmental well-being so so even that even there we have uh the concept of a fitness coming into this that is what out of all conceivable systems which are the most fit for purpose and now now soon we can talk about what purpose might be but you know that that is a new question that's that is a question that's hardly been asked in the in the history and i think maybe um it's only now that science has the tools and the theoretical understandings that maybe it's maybe this is right maybe maybe it is time that we can talk about the purpose of a society and how fit a given system design might be very interesting because in active inference land we're really familiar with talking about policy selection but we're coming out of the angle of model policy or simulations policy but that kind of question about policy setting and then sense making again maybe different groups use kind of different terms but all that sense making and problem solving has been siloed and the fact that there's not connection and common frameworks to bridge as you're placing it in one integrated brain like societal systems as a cognitive architecture that's not gonna work if the different sections are not properly having their within

and between connections working and we're seeing all these different sectors all these little regions of the brain health governance political legal justice education scientific analytical economic financial monetary you could go to a new site and on any given day see all of these things changing so very prescient to think about how the total system is going to be changing and finding new stable states otherwise it's going to be just on a spiral probably not in the right direction right as it seems to be unfortunately yeah um yes so so so the idea is can we design societal systems like economic and other systems such that the the set of them the set that is the cognitive architecture for for society can we design them so that they they are serving the same purpose that are they are they are integrated not separate systems and i think you were you were sort of referring to that a second ago but this the idea here is an integrated set of systems that serve a common purpose and for which a fitness you know a fitness evaluation a fitness score can be can be made is that something that we're going to return to is like how do we define common purpose how do we make yeah yeah yeah we're going to get we're going to get we're going to get into that okay just sort of doing some warm-ups here there's you know unfortunately there's the i mean maybe somebody might do it might help to read this whole series twice because the first time you are like getting just the kind of the big picture context thing like oh what's this what's this what's this then maybe on the second time around it's a little more clear like ah this fits in that's just talked about in the first paper but it really fits in in the third paper so there's just so many there's just so many concepts here it's a little it's a little chaotic at first as we cover some of these things right but the framing of the problem is not dissimilar then any number of other um perspectives it's really synthesizes a lot and there's a ton of scholarship in the paper so really i would recommend reading it before um imagining that it does or doesn't say something yeah please yes so i get that a lot yeah oh you must you must have meant this but i'm not done with the paper just fyi um here on slide five there's the sciencex link that people could click to and what is this about or what was this um showing uh yeah this is just a short little summary article that the science sects published of mine and maybe i don't want to be boring him but i might want to talk about these seven points this these seven points on the right there are actually a summary of sounds awesome because people who might like to read it or hear it both modalities and you're really going to unpack it in a new way so no worries about even stating it the way you stated it okay good all right so these are the seven main thrusts of the series so this is this is the spoiler alert so yeah so number one uh societal transformation is necessary if we're to avoid catastrophe catastrophe and maintain and improve social and ecological well-being that's the starting that's this that's

where this whole thing starts that's something that's transformation is necessary number two uh one kind of societal is is a science driven transformation you know you can imagine all kinds of there could be war uh revolution could be a a type of of transformation and i'm not talking about revolution here i'm talking about a science-driven evidence-based uh development of and migration to fundamentally new systems so we're talking about *de novo* *de novo* design from scratch right this is so we're not improving uh capitalism for example or represented democracy where we're looking at conceivable *de novo* designs that might be fit or among the most fit of all possible designs number three is uh now um you know if i were a genius which which i'm not but if i were a genius and i came up with the greatest plan that everyone could you know we could rearrange society according to this you know this design uh if there were no way to a practical way to implement that then i would have been wasting my time so a big part of this series is actually especially paper number two is really focused on what what is a how could this actually be done in the real world how can how can you do this um so i i claim at least that there is a viable and affordable uh way to go about transformation that within a reasonable span of time uh and i and i when i consider 50 a 50-year program here to be a reasonable transformation could spread to near global levels so um you know we're talking about a concerted effort over a long period of time to reach a global scale change but that does not mean that no change happens until the 49th year it means that change happens exponentially fast so maybe in the first few years there's you know there's not a lot going on but it goes exponentially fast from there those communities local communities that participate in this r d program would be obviously be the first to reap the benefits uh number four the and this is maybe one of the key world view aspects of this paper number one is all about world view the proposed program views society as a cognitive organism and its architecture so you know that if indeed society is a cognitive organism and ours and our systems are part of the cognitive architecture that already lends itself to ideas of how you might measure fitness now you're starting we're starting to get an idea of what is a system supposed to do so we'll be getting more into that today number five the intrinsic purpose of a society now obviously if we're going to build a new system we have to know for what is a new system supposed to do like what is an economic system supposed to do what is a governance system supposed to do what is the what is the purpose of them so uh uh purpose is also a big part of the world view in the first paper and the i one of the points of the fifth point is that the intrinsic purpose of a society of societal uh cognition and thus also of societal is to achieve and sustainably maintain social and economic ecological viability and vitality probably broadly defined now if you're

listening carefully and you're of the active inference persuasion you'll you'll already see active inference in here when i talk when we talk about um sustainably maintain that means and anticipation of the future all right uh i also say here the cognition is largely focused on reducing the uncertainty that our intrinsic purpose will be successfully fulfilled now and in the expected future so uh again we have a concept from active inference that is uncertainty um the cognitive view opens up many new opportunities for research and and i feel like this view is really critical if we are to truly uh have some kind of of optimally beneficial societal systems and uh number seven the last one this proposed r d program like i already mentioned it represents uh it's conceptual now but it represents a partnership between local communities and the global science community um and the you know neither of those are monolithic the global science community is you know it was a whatever however you might want to envision it a hundred labs or a thousand labs around the world or individual researchers or groups of researcher teams interdisciplinary teams at one institution teams across institutions that is what i'm that is really who i'm speaking i'm in the in this series i'm speaking to the science world and i'm suggesting or offering or or you know hoping that the science community might find this perspective interesting and see the the benefits that would be the scientific benefits that would come of this the the research gains that would come of this the possibilities that would come of this and and get engaged right so it's kind of a like i'm asking the science community to get engaged in this problem in it and in a particular kind of way and in what some people have called a second to you know whatever the phrase escapes me in the moment so i forget what the uh what is the title their second second i'm gonna just call it second order science but i think there's a slightly different phrase very interesting i remember reading this second order yeah yeah second order science so what what is second order science and i'm sure we're gonna again go into it in this like fractal convo but and how does everyone play a role in it right so so traditional science would be first order science is the the scientist is removed from the experiment that is an observer to the experiment maybe the scientist sets up the experiment but then steps back and watches the you know the results unfold and it just observes the results and then you know does the statistics or whatever on the results and then records the the uh the knowledge gain the aim of that is to build knowledge and you know that's wonderful and fantastic and terribly useful and important and that is mostly what science has done over the you know over the last few hundred years or so uh second order science is a little bit different it is the science i would say that is really appropriate for for today for our problems today in second order science the the aim is to change you know change the system change the it's

more about unfoldment of an experiment rather than getting to the end a certain goal of an experiment the scientist is part of the experiment it's reflective in nature it's intrinsically reflective and it and it really represents a partnership between the science world and and the stakeholders so it's a little bit more like engineering in a sense as more of a flavor of engineering in that there's a group of stakeholders who want some you know want something to happen and uh the science world is engaged in building that but it's part it's like a you know series of learning together and reflecting on what is important and what is of value so it's very much value driven which which is unlike first order science it's not particularly value driven so the the the aim is to change and it's value driven and the science scientists are part of the experiment really interesting that reminds me of of working uh for the first time really closely with someone outside of academia and then their insistence to include stakeholders in the conversation very early and that was something that i didn't see as a practice around me in academia so i didn't see it as an affordance i didn't see it as an opportunity in my projects and then when i kind of was like well i could contact anyone early on potentially and maybe even include them in the project that's um then you realize how closely clustered all not available have been organized and so it's really interesting and uh and as you pointed out like there still is a role for the first order of science it's sometimes you need to make the measurement like first order cybernetics the thermometer we need to keep the temperature range second order is it the right thermometer you know third order should we even be tracking a thermometer there's a space right for all right what is what do we really want what do we really value yeah and and maybe i should just inject here too that that you know it if if the science world is going to be involved in in a value question it really just has to be like transparent out front like this is what we're doing as opposed to more uh you know obtuse we're actually making value statements but we're sort of hiding them in first order science language you know so it just has to be transparent this is what we're doing this is what it's about we're we're learning together we're evolving solutions to real problems that's that humanity has and uh everything is like transparent and clear and above board and if it's done that way then it's i think could be extremely useful and it's is just what the world needs actually because there are serious serious as we'll talk about you know like overwhelmingly serious problems that we face i don't know how or if we can solve them and if we're not engaging you know the scientific world some of the brightest minds on the planet if we're not engaging them then you know we're we're not going to get the kind of solutions that might be you know i mean we're better off engaging the science world in these questions along with communities right because



that yep i i was talking to you no no no it's fun that's fine this okay yeah yeah okay just one short on that is it's actually a really respectful way for science to approach this question of like we'll be ready as scientists or as systems thinkers or as modelers will come to the table of policy and value discussion when we have a transparent model that's actually going to be adding value rather than potentially being a vector for some non-transparent force to have undue influence on discussions that really matter so it's it's um i hope this conversation helps people rethink how science could be involved in these policy discussions because those are going to be the exact terms that we're going to be talking through so right right and when we talk about the second paper we'll get we'll talk a little bit more about uh second order science and its role and there's citations there for anyone that's that's interested nice so here we go to number six why transform yeah so uh i mean i i'm guessing that uh for your audience you know the the the your audience probably has a general sense of things are not good you know things are not moving in the right direction and i think maybe lots of the public has the sense that things are not good not moving in the right direction and climate change is certainly part of that bio on the ecological and the ecological theme climate change and biodiversity loss are perhaps two of the you know biggest challenges that humanity faces uh you know now and over the next say 1 000 years or something but but our problems go deeper than that it's it's not just that there's that the temperature is warming it's not just the the birds are dying and the insects are dying it's that it it's also has to do with social issues poverty for example uh financial instability uh uh and all the all the social problems that climate change and biodiversity loss and other things are going to to bring uh uh already with covid i think there's there's the the uh split between the rich and the poor and the world is getting even larger than it was and was already disgustingly large there's a question of power you know why do why why does it in a sense if we think of money as a voting tool why is it that a billionaire has a billion times more power to decide what society should be like than i do you know like there's questions like that there's just fairness questions of fairness there's questions of decision making who who how do we make who actually makes decisions like how how many people are in like how much of the of society is involved when society makes a decision you know there's all these questions uh and all of them can be i think addressed together uh in a in a r d program like this and we have to we have to address them i don't think there is a there's an option of not transforming so so voices for transforming for some kind of bold transformation are getting louder and louder both in academia and you know at the at the global scope global policymakers at the u.n and other folks like that uh just within the general public with even

within some uh you know municipalities city states local governments and and even the national gov governments there is a rising rising chorus of voices saying we need to transform and we need to go big and we need to change things you know kind of radically somehow so that we can get what we want in the you know like so 50 years from now my children can be enjoying a better world not a not a dying world now that doesn't mean that i'm going to jump just a little bit ahead here that doesn't mean that everyone thinks that transformation is necessary but lots of people don't lots of people would just as soon keep everything just the way it is you know or they just haven't thought about it maybe or whatever so so i just want to make clear that i'm not you know it's fine people you know people can fall wherever they fall on the spectrum but the paper the end the r d program is really aimed at that percent of the you know the population and the science population that is interested in vole change and it doesn't need the r d program really doesn't need the rest of the world to even engage or pay attention to this the this whole this the r d program is designed to be successful even if a small slice of the population participates so so we you know that was a long story to say that we face severe social and ecological risks and they're getting worse i do want to mention one more that kind of second secondary risks that we will be facing more of already we have migration forced migration problems people fleeing lack of economic opportunity and fleeing violence and things like that but you can imagine that that if that's happening today and climate change hasn't really even hit yet and biodiversity loss really hasn't even hit badly yet or at least this hasn't hit widespread badly where are we going to be in in in 10 years there's going to be millions i read a paper and i'm sorry i don't remember the citation but i read a citation a paper that said that um you know possibly uh a third of the global population could be migrants in the future you know in the coming decades i mean that's billions of people have to go somewhere and we are not prepared at all like in america or really anywhere we're not prepared to absorb those people bring them into some kind of productive engaged society where we're all working together and cooperating to you know to to address the needs of we're not anywhere near that and maybe i'll mention one more too because it's one that people don't usually think of but the even if there's no uh even if there's no uh catastrophe even if things plug along as they're going and there's no mass die off of humans or anything like that the population is set to decline i don't know when the peak is supposed to come but uh the peak is supposed to come at you know within the next 10 20 years or so and after that the world population will start to decline how is how is this growth capitalism model growth-based capitalism model how is that going to function when the world is shrinking you know so

there's there's just there's there's there's short-term issues there's long-term issues there there's just i would say overwhelming evidence that what we're doing is not sustainable on any level and if we don't do something it's going to lead to to even greater catastrophe i have a few questions one just from the chat they wrote the science community here seems to be referring to the natural rather than the social sciences so it seems more like this is about stem and technology so maybe where do the social sciences fit into this oh i i in the in the second paper i give a laundry list of fields scientific fields that i that i think could really contribute to this project and it's essentially a to z you know from anthropology to uh zoology you know like uh every really every branch not and not only every branch of science but art uh you know media there's there would be room for every you know kind of uh branch of this human endeavor to get involved with this thing agriculture and psychology and public health and you know sociology and all that would just be like at the moment you know as i said this is not a funded program but i and if it were funded i anticipate that the you know either the organization would act as a clearing house for for funding uh for for you know proposals or or there would be some kind of mechanism set up to to distribute funds or or i'm not sure exactly how that would work but it would be my hope that funds are distributed to literally every branch of science and then beyond science to like i said to art and to literature and to you know all of that stuff well then let me ask one more question on this slide so the communities that are likely to institute for example the flood prevention some sustainable strategy they're going to be in a flood area or they're where the sea levels are changing for example so when we think about cognitive architectures what kinds of individuals or teams or communities or cognitively are like the early canary in the coal mine that you think are ready to transform or somebody who like might hear something about a system they're involved in and think actually yeah that sounds like my organization or self might be at this sort of transition point right there you know if if one looks around on the web and elsewhere there are numerous uh kind of experiments going on in in a variety of things in the and new forms of representative democracy new forms of decision making new forms of economies in the sense of you know local digital currencies and things like that i think all of those you know all of those are are excellent um you know a resource to draw from uh of the the task is then is to take these ideas these ideas that are springing up all over and put integrate them in a way that is functional you know can serve a community initially maybe the community is small just a few thousand but the idea is that it would that it would grow over time exponentially grow to who knows you know hundreds of thousands millions i don't know so how can you take all those ideas and

actually make them work sometimes i i liken it to you know that you might tinker in the garage with a with an airplane you know you might build a two-seater in the garage and that's totally cool you know you can you know that maybe what the wright brothers did or something that's fantastic but i'm really interested in building a jumbo jet that takes you know 500 passengers at a time in an hour and a half to from new york to london or whatever and doesn't fall in the ocean you know like so how do you do that how do you how do you build a integrated system that is safe that is resilient that is that it that has metrics that you can monitor progress that has good anticipation so you know where the you know that where you're going tomorrow you know where this where is this going you know what's going to happen tomorrow and sort of what's going to happen to me you know that's that's part of the question so uh so the so i think the challenge is to take all these ideas that are that are popping up all over which some really great ideas and then to integrate them into uh into a coherent hole that that spans every one of the i think maybe is it six systems that i talk about so that so that they're not designed in silos we're not just building the new economic system we're not just building a new educational system we're building this we're building a a cognitive architecture that includes all of those okay awesome response awesome response all right well we're you know we're moving slow here it's totally fine but uh uh so uh on this slide uh this is again again is the topic of why transform and we only got the first paragraph so far oh oh nice okay we will stay on six oh you know we and we gotta we got a ways to go on six we can uh we can continue on you you can just summarize in one sentence or um however you want to do it but up to you okay all right i'll try to go a little quick through the rest most of that discussion on is on humanity faces severe social and ecological risks the next one is our systems are dysfunctional as is and i would say that is evidenced by the fact that we are under extreme threat like you know what kind of sensible uh healthy person puts themselves on a ledge hanging by their fingertips i mean like that's not something's wrong you know like you don't do that a a competent a cognitively competent society does not push itself to what is could literally be the the early stage extinction you know that's there's something deeply wrong deeply dysfunctional that we find ourselves in the situation that we're now in and we can do far better and we must do far better obviously if we're going to thrive and survive and not only that but as we talk about purpose we we actually long we it would make us deeply deeply happy to do better to be to solve problems to be more engaged with a healthy society to to really feel like there's a deep cooperation happening in society that would you know our hearts would burst you know of of uh of joy sort of we would be we would be happier if we were a

part of a society like that it's very stressful we were designed by evolution through evolution we have become we were i really every organism as we'll talk about in a minute is a problem-solving organism and if i can't solve problems there's like a you know like fundamentally going against the grain of what it means to be an organism okay nice good shall we go on to the next one you can go to the third part or do you want to go to slide seven uh what is the third part well it was kind of the i think that was about it let's go to slide seven so john what is this series [Laughter] well i'm glad you asked oh i have a script don't worry so as i mentioned everybody the series really is a is a proposal for an rd r d program aimed at as new de novo development of new societal systems and it's also a way to context and a way to think about what transformation might mean so uh it is it is a long-term project you know like a 50-year project this isn't we're not it would be dangerous to change society radically overnight so this is a long-term project for long-term benefit and then early communities that become that become involved early in that process would of course see benefits quite early um and there are i already mentioned that we cover six primary societal systems so that's the once again they are economic governance legal public health and what i call analytical forecasting and education and you know any one of those can be broken down further like economic really me and also includes a monetary system and the financial system and things like uh so six primary societal systems the idea is to out of all conceivable designs what might work best uh the the whole project is based on three underlying propositions the the papers are and the project first is that uh a society of any scale and i don't mean society is in bill millions or billions of people i mean society as in a thousand people you know like a sub sub city a community that is not even a whole city just a a group of like-minded people uh who are willing to give this a give this a field trial ago a society of any scale can be viewed as a super organism so that's kind of fundamental everything really really works from there we are together we are not just individuals connected we are a whole society is a whole and it's a and it's a whole with the environment and it's wider you know sphere so as we'll talk about today you know this even the idea of an individual is it's okay to talk about individuals it's fine but it's kind of like an arbitrary thing an individual could be an individual cell or an individual person or an individual uh species or an individual ecosystem but it's all with all deeply embedded and enmeshed entwined with the whole so uh uh a society can be viewed as a super organism uh uh society's complete set of systems as we've already said the six six big systems i've mentioned can be viewed as a cognitive architecture it's it's the means by which the society learns decides adapts and and this society's efforts this is the third underlying position the

society's efforts to learn decide and adapt and be viewed as being driven by an intrinsic purpose and that's really key also because it's not just that we're learning deciding and adapting willy-nilly i mean i mean maybe it seems that way in the world you know in the sense we're so dysfunctional it kind of is billy nilly but but what really matters is that we learn decide and adapt in relation to whatever intrinsic purpose we actually have as as a society as individuals in a it's that it's it's it's it's as i will use the the term uh maybe several times today it's solving problems that matter that really that really matter that's what we're after cool well on this uh super organism point as someone who studied ants and um still studies the eu social insects this kind of multi-scale thinking is really fun it's just always such an interesting question what level does purpose exist at so is it that purpose exists at every level like the cell the nest mate the colony the colony and its symbiont and its predator prey and the microbes and the niche and then where you draw the sort of qualia like philosophy lines it's it's it's a whole debate it's a whole thing yeah it is it is it's rich it's a whole it is rich but then also there's the systems kind of systems mapping approach that albeit coldly does sidestep those and i think part of the humanism is about how to bring these multi-scale questions and multi-scale perspectives which are basically neutral tools because one's preference vector and active inference could be to lose money i mean then i guess you know good job but we have to have the preference vector and then what will be interesting to draw out i think in these discussions will be just like you said with the transparency of the values what does it look like to actually specify values within a model based framework and take the bold step from just like i trust my models and i'm a fan of liberty to i trust my model about liberty and i'm willing to let that model drive for a little bit with respect to my decisions so that's really like such an interesting handoff and that's also new and related to technology that's evolving so like you're bringing a lot of ideas together i hope people are you know listening to it and finding it exciting enable just jump ahead a little bit here as far as a preference vector i i took pains not to put a not to suggest a preference vector of any kind in this series this wasn't it wouldn't be very useful for me to do that for one thing but but but for those listeners who are active inference fans you know having only a preference vector is uh is or maybe i should say having a having a checkoff list like you know there should be this level of education there should be this level of [Music] health people should live this long and so we have our fitness and we're gonna uh we've decided in advance even before the system is running we've now have a list of things we're gonna check off we're gonna score each one we're gonna come up with some kind of integrated fitness score from that and that's how we're going to move forward we're always

going to refer to this fit this you know this fitness model and the fitness vector and these and these kind of hard-coded values for what's good and what's bad so in the world of artificial intelligence and in the world of active inference you know that really doesn't go very far that doesn't work that doesn't work very well because what happens is we didn't you didn't think ahead you like you some something happens tomorrow and whoever came up with that list of uh you know those values or that model didn't really include the fact that maybe spaceships from mars were gonna land and cause a new disruption and then we have to deal with that problem now too before we deal with anything else so that wasn't in the you know that wasn't in the plan and now what do we do so there's right so so this is you know this is really where active inference plays into that's one way that active inference plays into this is how do you evaluate and act in a world that is full of uncertainties right the unknown unknown the unknown unknown is the temperature dynamics but you know it's going to be temperature and so how can you plan for what you know it will be in a distributional sense right and make stabilization on that awesome right right so so yeah so you so you realize you know already you realize maybe that this is not a proposal to build a say like a model of uh of uh you know like how society makes decisions you know that's that's not that's not it it is what is the process by which society cognates and you know what kind of what kind of infrastructure and tools and and and you know mechanics can we use that would facilitate that but it's not to build a thing it's to build it's to realize that we are in we are engaged moment to moment in a cognitive process society as individuals are and how can we do that together as a society so that we're you know we we balance exploration with exploitation um you know so that we we learn about our environment we grow we learn we explore we we make good decisions based on available evidence and based on knowledge based on cultural knowledge you know like all those things right so so this is a this is the the the you know i think organisms are a process they're not a thing anyway right cognition is a process and societal decision making is a process and really society is a process you know there's there's not too many things in this world there's mostly processes living processes intelligent processes so that's that's the that's the hope that's where this is trying to go is to like with that in mind with that with that broad understanding or broad concept in mind how do we uh how do we think about you know how we how we come together as society how we cooperate how coordinate how we make decisions how we how we learn how we explore what do we what do we monitor what kind of information do we seek you know what kind of experiments do we do all that kind of stuff great do you want to go to that seeks answers to two questions or yeah yeah that's the last one

on this on the slide is uh so so the two questions that we hopefully would uh try to answer with with this r d program is and and one of this i already mentioned but out of all conceivable designs for societal systems so so so this isn't about capitalism versus socialism or something like that there's like i would think there's an unlimited potential we're creative we're creative people there would be a million varieties of of societal systems and integrated societal systems that we might come up and some of those probably would work very well and some of them probably would work very poorly um so among those what what might be among the best and not the the single best that's not the purpose either it's not just to find one thing that works is to find like a you know more of a a variety a process of things a mix mishmash of things that community the communities can choose to implement that you know works well for them and that suits them and that works well for their neighbors and works well forever it works well for the whole really uh second question is uh by what and this is as of course the the kicker is not only what systems might be best but what how would you possibly uh implement them what viable way is there to implement new system designs you know like otherwise what's the purpose you know this isn't this isn't an academic this isn't just an academic uh you know excursion this is like let's change the world and that has to be practical in some you know in some has to be viable so those are the two questions what what what might be best and how do we get there awesome well it really does i think read like you've turned it over a lot and communicated with a lot of people about it so yeah we can go to eight should we go to the next one yep slide eight okay so so let's suppose let's suppose your listeners are with me and you know we kind of agree like okay yes transformation's necessary and uh again i want to emphasize i'm not talking about reform i'm not talking about a softer better capitalism i'm not talking about you know improved voter registration or like any of those things i'm talking about de novo starting over from scratch what might be best and if it turns out that the old systems were better than anything that humanity can come up with well then you know that's the answer but i can't imagine that's true because the old systems were never designed in any kind of you know thoughtful science driven you know process to to to test to explore and to come up with fitness like what is the you know we don't even have a fitness for our current society much less of fitness for societal designs i mean we have the gdp but that's a terrible terrible limited fitness metric okay so suppose you're with me suppose we're we're on board we we want to do this de novo design thing where do we start what's the what's what where do we even get off the ground on this and i suggest that the way to do it is through first address worldview from world view once we understand what the world view is what a reasonable



useful world view will for this project then then purpose derives worldview begets purpose once you understand what it is you want what you value what do you value once you understand what you value then you can say well i value a and therefore the purpose is to have a manifest in society for example so once you have purpose then you can think about what metrics how would you measure whether are you so here's a new design is it fit for does it do does it fulfill its purpose you know that's the question and then metrics go with some kind of fitness evaluation and then finally last of all of those would be the design okay we know what we know what we value we know what this thing is supposed to do we know what the purpose is we know that attractor is supposed to you know plow the ground or something we know what this is supposed to do we know how to measure success and uh now finally then let's talk about design what are the what are the you know the specifics and mechanics and how does that happen and the the series is really kind of laid out this way the first paper really talks about world view and purpose the second paper talks about the you know the more the mechanics of things like viability how would you make this thing viable things like that and then the very last paper that's titled the subtitle design okay so uh that's how we uh and and maybe i will just mention here that i put metrics before design because we might have some ideas uh getting back to that preference factor we might have some ideas like we would like people not to die at 30 you know we'd like people to mostly live to a ripe old age and have you know enough water water to drink and food to eat and all that kind of stuff so uh you know what kind of design once now that we have metrics to measure that kind of stuff longevity and nutrition and things what kind of designs would help us to reach those targets you know so that's one reason why design why metrics comes before design okay so uh now we can start to jump into you know now we jump into this topic of world view the very first one world view and purpose and uh um you know i think we're on page 10 maybe of the first paper so far we've still got a long ways to go nice uh okay so um so what what i've tried to do in in in you know the main theme of this first paper is i've tried to lay out a world view that is cognizant of that reflects some of the latest developments in in science and in a variety of fields and sciences in those fields would be like complex system science cognitive science evolutionary biology uh in a you know a few fields like that information theory and a few things like i i've tried to to outline a a a world view that makes sense from that leading edge of science and i would say too that that science has gone through really kind of a revolution you know there was it's kind of like there's the pre-1950s 60s science and then there's what we have today and there's enormous jumps enormous leaps in understanding that have happened just in the last

say 50 years or so and some of those leaps the ramifications are only now being you know the they're now being felt right the the the the some of the concepts are a a distinct shift from where we were in the you know the pre-1960s or pre-1970s or so and um obviously we there's also you know we we see the changes in our lifetime you know like we i was watching an old show on tv the other day and somebody put money in a pay phone you know like they put a quarter and a pay phone to make telephone call and it's like okay well that's that's that's history you know so there's all these technologies that have that are that are that i grew up with that are not they don't even exist anymore they've been replaced by entirely new frameworks and that's the speed of those of the speed of that evolution is is exponential so it's tremendous changes happening very quickly and the task of the program oh yeah which one put on that would be um potentially because you're taking such a broad perspective with complex system science and evolutionary bio you might say that society has always been a cognitive architecture but if you had asked in 1500 is society a cognitive architecture be like well no i mean you have agriculture you have this you have this you have that whereas now if you tell hey telecoms are they run through everything and the internet of things the internet people you know like all this sort of stuff you tell people actually it's a multi-scale cognitive architecture humans are in the loop and our algorithms are never independent from us they're in feedback with us it's like yeah that was what the mainstream was telling me so actually it's a total alignment point because it reflects how rapidly things are changing that it's just undeniably obvious that the communication infrastructure is the system that we're engineering right right absolutely yeah yeah communications have mind-blowing changes and communications and that brings mind-blowing changes and outlook but but i want to emphasize a few points to this worldview that that you know it's not it's not just that everything is connected like you know you go like i know what a complex system is it just means everything is connected and we're all kind of whole and blah blah okay fine but but even for people who are in that ilk you know who understand the the basic concept there there's ideas that are that have come out in the last decade or so that that are there that are even pushing that boundary you know right and and i just want to highlight a few concepts here and i think active inference really is playing a you know is is like a in a sense a culmination of some of some of these ideas or an embodiment of some of these ideas the main thing i want to say is that life is intelligent and whole so it's not just that everything's connected it's that everything is intelligent everything is a lot you know life is an intelligent information processing thing everything is is is adapting learning deciding whether we're talking about everything is cognitive you know and cognition really implies

information and information processing so whether we're talking about a slime mold or a human you know there's there's in in everything in plants in bacteria in mold and anything that has any life at all that can be considered alive is intelligent and is learning and reacting not just reacting but learning reacting and also deciding and acting and remembering and all those things and you might ask well you know that's impossible bacteria doesn't have a brain you know it can't be it can't be cognitive but it is cognitive but we just have to relax what we how we define cognition you know and when on the slide i have a little thing there that every organism is cognitive in the sense that it displays capacities typically associated with human cognition such as sensing learning problem solving memory storage and recall anticipation anticipations is key to everything and attention is key to so every organism does that plants and everything else and it doesn't require a central nervous system and and you i might add to this that not only is every organism cognitive but essentially every organism organism is cooperative to those cooperation and cognition go hand in hand because any intelligent any organism that can act to better its you know viability is going to cooperate meaningful ways with other organisms and you know other species and things like that nice point because um there's cost to communication whether it's exactly whether it's the cost of making the pheromone or just the time which is super finite or attention fundamentally and so costly interactions through time the game theory are either to exploit and stabilize which is fragile or to succeed together yeah exactly and and and succeeding together cooperation is is like everywhere once you once you understand what you're looking for it's in the biologic world it's like everywhere so this idea that we're you know one one one person against all or you know we're a dog eat dog universe i mean it's you know in a certain sense it's true obviously tigers eat you know whatever they eat zebras or whatever i mean that happens yes of course but in the larger picture over and over multiple time scales not just uh you know in five minutes but over evolutionary time scales and uh you know developmental time scales and everything the cooperation is really the rule for the most part and if you need if any listener needs proof of that just think of who you think of your body i mean there's about a trillion some trillion some cells that are enormously harmonious like your blood pumps every day or you know this is a this is like a miracle i don't want to use the word miracle because i want to get into whatever that might imply but uh it is amazing aw inspiring the the depth of cooperation just in our own bodies is like that's that's like evolution must prefer cooperation or else there would never be such a complex uh pattern of cooperation as we see just in one human body just to give one example from the bees so from a species i study it's almost like a

sparring type of cooperation because when it was discovered that there were some workers with developed ovaries there was a whole story about cheating and policing and about altruism and this equation says this and that equation says that and then when you take a step back it's like the colony having a distribution of over-reactivation may be more ecologically resilient so um i as an evolutionary biologist never think well my interpretation of what would be lovey-dovey in this system must be how it works because that's so clearly not true it's just to say that there are interesting dynamics within and between levels and in the long run cooperation and stable cooperation and like learning to adapt to your niche is a winning strategy in a way that locking down just isn't but unfortunately under high um stress and uh high uncertainty conditions simple strategies can become rife so that's sort of a failure mode of the population yeah that's what i understand also like you know what you're saying yeah no it's uh nice if a if a species can avoid highly stressful situations because often things don't go well once you get to that point you know things can go awry what's that do you want to go to nine or stay on eight oh um uh yeah i think let's go to nine let's move ahead perfect what is an individual okay so why why the why in the world would i why would we ask this question and why would i spend you know multiple pages in this paper even discussing like of course we know what an individual is right or or maybe not like like that actually turns out to be a difficult question what is an individual and it's important to this and it's important to this discussion of societal systems because who are we who what you know what is the purpose of a societal system what is it what is it who is it supposed to serve you know so you have to ask really like it's it's good to ask if we're going to build a societal system who wh who is it that it's supposed to service you know like who are we what do we want you know it's part of figuring out what do we want what do we value who are we start there you know i would say so so we've already kind of touched on these themes but this idea of rugged individualism you know like from a certain perspective and a certain you know from a limited sort of time frame perspective sure there's there's a rugged individualism that exists right and it can be useful in certain certain situations but by and large that's not what life is doing you know that's not what the the they're um we are we are it's really even difficult to say like where if i'm a rugged individual where do i actually start and where do i end you know like where is where is me this you know even physically it's hard to say because this physical me is really i think more bacterial cells than it is um human cells right so so uh like i'm a sieve i'm a i'm a process through which things are flowing through i'm a i'm an ecosystem myself with bacteria and viruses and human cells and all of those components are necessary for me to survive today and for for humans to survive

you know over eons were like a mix we're a bag of of human-like things and bacterial-like things and viral-like things and and we're porous and we're part of the carbon cycle and we're part of the nitrogen cycle and then you and then when you say like okay well how could you be a rugged individual individual when you're really this this porous smorgasbord of things right so so the there was an interesting paper that came out i cited in the in my in my in paper number one that uh was looking at this question of what is an individual and they were looking at it from an information theory standpoint so they came up with this they came up with this uh uh theory uh and i think do they have a name for it yeah uh information theory of individuality and they say base it's done at the bottom of the slide there and they say basically that uh you know an individual is a process just what's what we've been talking about before that propagates information from the past into the future so that you know implies uh information flow and implies a cognitive process uh it implies anticipation of the future uh and it probably implies action and this thing that is an individual it is not like it is a layered hierarchical individual it's like you can draw a circle around anything you know in a certain sense and call it an individual under you know with certain uh definitions you know if you want to define what its markov blanket is but uh but you know we are we are we are our cells are individuals our tissues liver say is an individual um a human is an individual a family is an you know and it just keeps expanding outward from there the society is an individual so it really it's none of those are have you know any kind of inherent preference levels there's no preference to any of those levels everything's an individual layered interacting overlapping individuals and it's just it's just a it's really just a the idea of an individual is just where do you want to draw your circle and then you can you know then you can talk about an individual at whatever level you want so so that's all about information so it's all about processing information right and that's and so society is an individual and we are part of society and we're talking about societal systems and so at that that seems to be our level of focus here at that level we can talk about society as an organism as a cognitive organism that is propagating information from the past into the future and from an active inference standpoint we can say uh under the you know under the fitness score whatever word you might want to use for that of reducing uncertainty so we act to reduce uncertainty right one interesting point on the ant colony again that's just it's the example i'm coming from is the 1911 paper by william morton wheeler is called the ant colony as an organism not a super organism because superorganism implies that nest mate would be the true and right that's something that insect with a six-legged version is now it's the whole super organism oh well the ant colony is society that

whole frame is actually the shadow of what the evolutionary reality is which is that the ant colony is an organism not a super organism and the ants are tissues and so which level we prioritize or do we say no there's no a priori level ant is just i'm not even gonna say there's anything out there called ants it's it's you how you're thinking about it or do we get lost or are we going to find a ladder in that multi-scale yeah well the latter is you know the latter is active inference because it doesn't say active inference doesn't say make a make an internal model of the world that is accurate that actually accurately captures all the all the details of the world of the universe that's not point that's not what the mind does that's not the point the point is to act under uncertainty given some useful model of the world act under uncertainty so that your fitness score improves and by fitness score here we essentially mean you know and anticipated uncertainty so i i would very much like to be have some certainty that i'm going to be alive tomorrow and if it's freezing outside and i don't have a coat on uh you know that that becomes iffy so uh i'm going to be happy if i'm going to be i'm going to go find a coat because it is going to reduce my uncertainty about survival over the next 24 hours but you can expand that you know outward right we we need to act the all organisms are acting under and and we can think about that as we can think about that um we can think from that perspective as a society of what are we doing and how do we measure success well we're measuring success by acting under uncertainty and then re and then paying attention to what happens and then acting the same or differently or you know some other way or somehow some were then choosing to act again in this cycle of act uh you know act uh process act process act process you know model act model act model that reminds me of course of the ooda observe orient side act model and other sort of cyclic models of action and perception and then i would say that active inference provides a few nice little benefits over other phrasings of action and qualitative and philosophical ones like inactivism as well as quantitative ones like cybernetics and other kinds of control theories so i totally agree this is nice stuff and uh nice we can carry on if you want you want to go to town okay yeah yeah ten let's go to ten so everybody's listening you can ask questions in the live chat or you can um join us in the jam board and flip over to slide ten so uh so so i so i'm thoroughly enjoying this conversation but is it true that we have about uh a half hour left of it yep we have 30. is it going that fast okay i believe so we can we can slice slides or move slides to another day so however you want to do i'll i'll try to go or i can chill for a few more minutes afterwards try to go a little quicker but there's so many things to say you know there's you know it's it's one thing to read the paper and you know kind of written in a you know sort of scientific terminology and you know it's but it but it doesn't really convey

the feeling you know you have to kind of think about what is the feeling of this whole project and we can do that better in a conversation i think agreed okay so so so then you know we're on this topic of what is what is our worldview what do we value and what is our purpose and then we've come to this question then okay so who the heck are we then you know we're we're and it and not only who are we but who are we building these systems for you know what what what is what should societal system serve who or what should societal systems serve and the only reasonable answer that you can come up with is that societal systems should serve the the extended self like not just the body not just the family not just the you know the thousand people in a society or the ten thousand or a million or whatever but their environment the the society next door that they're engaged with and cooperating with and coordinating the society across the planet that they're sharing information with and learning together with and so it's the whole that we are metrics as we as leaders who come to metrics those metrics have to represent both the cognitive process how good how are we cogniting how well are we cogniting are we functionally cogniting and are we achieving through that cognition are we achieving the kinds of aims that is serving the whole is the environment improving is the you know quality of air improving is the quality of life improving for individuals right um yes so uh we are so this in a nutshell we this is the world view in a way we are in intimate with our greater world we are individuals but of the nested overlapping variety individual cells bodies groups communities ecologies nations and all of civilization we're not separate in any absolute sense and there's no privileged level or scale to any of that nor are we passive bystanders in this unfolding this is not this evolution is not it's just a chance thing like by chance somebody does this one day and then evolution goes on another another avenue no there there are there are opportunities in the environment uh that we can react to that lend themselves to to providing information or providing gain of benefit of some kind and and you know we are driven we are are we are consciously creating and you know even a really great societal system that integrated societal systems would be consciously creating acting cognitive acting cognitive and consciously creating and it towards some towards some goal and that goal then has to be you know the maintaining of vitality being the in the for the extended self all right so that comes the slide that's that's the topic of slide 11 and what is our purpose so so so over on the right there i just want to reemphasize we are anticipatory we are cognitive we are problem solvers we are a we and then i have below that i am a we you know like i i am i can i am i'm intimately connected with this i'm i'm everyone in that sense you know yep yeah the whitman um you know i contain multitudes and also gilbert at all i have a paper called um we were

never individuals kind of on that wavelength that you were talking about with the sort of distributed systems all the way down approach and also dennis noble no privilege level of biological causality similar uh basically realization that perspective complexity science basically entails either the choice of a priori level like saying it is multi-scale and humans are the best scale or gaia is the scale or quantum is the right scale that's a claim as well as it being a claim actually there's no privileged level of causality so that's the sort of table as it's said right right right right right and you know what it's not that really this this entire project you could say like a sentence you could say this whole project is to help us be who we are more be more uh honestly who we are more real to who we are right it's not the it's not to to have people behave in some unusual way or some altruistic way or anything like that it is it is to have it is to be more more ourselves more fully ourselves more completely ourselves and then all of these pages all these things we're talking about is who that self is who who are we really and it's about the adjacent possible for who we are who we are is not an essence that is uh there's uh seven seals and it's being unlocked it's actually something that's being drawn out through inactive realization in the niche through niche modification through stigma through becoming and and then the adjacent possible is where the imagination and the planning comes into play and if people are hesitant to talk about the adjacent possible for who we could be just think about chess it's the adjacent possible with the strategy on the board and we're talking about the adjacent strategy possible for who we could be in terms of our strategy for you know all these recursive layers our strategy for how we think of ourselves and all these other things you're talking about absolutely absolutely and then and then ultimately serving the serving the kind of fitness purpose of you know if we take action a is that going to reduce our uncertainty about those things that we that really matter you know that are that are the the the key variables you know okay so uh so uh you know this is maybe summary now we we've we've talked about about about who we are and i just want to say a few words then we have a purpose and uh from like a biological i call it an intrinsic purpose but like from evolution by being the fact that we are a part of life we have a purpose because all organisms making capability casual power causal powers and the intrinsic purpose of an organism is to achieve and maintain vitality a sustainable flourishing of self which can include that extended self and we do that by sensing and evaluating states of the world and ourselves and implementing appropriate actions that that are based on anticipation we we anticipate what will happen if we do or don't take an action and we choose if we're for functional we choose those actions that can serve our intrinsic intrinsic purpose of remaining vital into the future so anticipating



vitality and that obviously implies some kind of modeling of the world anticipation implies some kind of modeling in the so that's an organism's intrinsic purpose and then society by you know by uh uh you know it's just that's necessarily shares a similar related intrinsic which is to achieve and maintain vitality maintain and maintain and by maintain i mean anticipate into the future maintain vitality which is accomplished through cognition and cooperation so the self that we must keep vital is the extended self and it follows that the intrinsic purpose of societal systems like financial systems and other is to serve the intrinsic purpose of society so now we know or you know i mean we don't know i'm just i'm just putting this out from as my take on it but this is what i'm offering as a concept for the world to chew on you know uh all right and obviously i'm getting these ideas i'm not coming up with all these ideas myself i'm digesting hundreds of other papers that have been put into this kind of submarine in a way um so that's uh so that's why we need to our purpose is to remain vital into the future and and that and when we talk about the self that remains vital it's the larger itself all right so um slide 20 uh or excuse me 12 yep yes slightly so there's a lot of discussion about complex systems you know we've been discussing complex systems and i just want to make a couple of here because uh commonly some it is not uncommon that someone will say a complex system well that just means that it's liable to fall apart at any moment you know it's just too complex it's going to crash uh but and that that obviously can happen you know systems can collapse quite quite true but obviously life would not be doing very well if the if the evolution builds complexity in species and you know in organisms and ecosystems if life would be have a rough go of it if it was so fragile that uh complexity became a burden and and uh you know come and then you know you reach a certain level of complexity and then you fall apart that's not really i don't think i mean that can happen but that's but but complex useful complexity doesn't make you fall apart it actually just does the opposite it serves what we've been talking about all along and that's problem solving so we are anticipatory organisms we are problem solving organisms it's our nature most of what the human brain does is to solve problems of one kind or another social problems physical problems whatever and maneuver in the world you in a useful way and complexity is what allows that there's a number of studies that i cite here that show that as an organism even as a robot you know faces uh more difficult pressures from its environment it complexifies and complexifies by complexity then it's it's it implies a greater number of parts coordinating or cooperating in some way uh to you know solve this new challenge and obviously as a human we're very complex we have we have complex needs we have we can think not just what's going to happen in the next millisecond but what's going to happen we

can think about what's going to happen in 100 years i mean part of this project is to think about what might be happening over the next hundred years or even a thousand years so as an organism complexifies it become it at least potentially becomes a better adapted to solving more complex problems so you could and from that sense you could almost ex equate complexity with problem-solving capacity you know at least in a uh you know in a general sense and then i talked about well that just reminds me of in the free energy calculations that we um have gone over in various papers it's accuracy is the modeling imperative and then complexity is tolerated to the extent it facilitates accurate modeling so if you get the one parameter model and you got 99 and it's adequate and it's good then you're good to go and you're gonna go for simplicity but then what you're saying is actually the um appearance and the hallmark of complexity in the world it means that that organism has the need to solve problems at a given level of counterfactual depth or inference skill or temporal depth temporal thickness exactly exactly exactly yeah that's that's it yeah yeah so uh so i talk in the in in this kind of middle of the paper now we talk i talk about a few ideas good regulators requisite variety self-organized criticality and then the free energy principle from active inference um and uh maybe i'll just try to briefly talk mention what's what those means for what those ideas mean for people who aren't familiar so good regulator really came from the good regular theorem or whatever it's called really came from cybernetics ash ashby yeah a lot his law of requisite variety and uh the it's the concept is that a organism or a you know a system must be a model of that which it but that needs to control so so i am humans are a model of their niche their their physical niche the gravity our bones are a model of gravity or you know we are our our our our complex problems that we face of maneuvering the world with our senses and our sensitivities and our uh uh our you know vulnerabilities we have to be complex enough ourselves to handle a complex environment so that's just another way to saying that that systems complexify in order to handle uh you know adequately handle their and and part of that complexity then is having enough dials you know the system has to have enough dials and enough levers and enough movement uh opportunities to control that which uh to control that which they need to control so a good controller has a similar variety to that which controls that's the requisite variety part and i just the technology allows us to um sort of play with that for example someone driving a car with the affordances of just their arms and degenerative model of the road can be driving in very challenging situations so this doesn't mean that you need to have the road inside of your head or need to be the road to drive on the road it's a statement about how action-oriented choose actions right right and maybe i'll just

mention that that had that that plays out on all time scales right so the immediate mechanistic time scale the developmental time scale you know the uh the uh evolutionary time scale and everything in between right so you know because like uh an organism you know think of an organism that only once every century that has to deal with a a you know a one-year drought or well somewhere in the mecca in the mechanics of that organism has to be a little piece that is capable of digging down into the dirt and just hanging out for a hundred for a whole year or whatever without water even though that only happens once in a total blue moon you know like like we have to have this we have to have this flexibility within us even for those extremely rare but deadly you know potentially deadly uh scenarios and situations that we might face right okay so complexity essentially you can almost equate it with problem solving capacity and again the world numbers uh slide 13 513 help organize criticality yeah so so maybe i want to just kind of back up for a moment and just say like again what is the purpose of going through this stuff you know i i mean i mean it's useful sure if you want to build some new system it's nice to know what the purpose of it is you know i mean obviously but uh but these ideas that we're talking about and self-organized criticality is one criticality is one of them these ideas can not only serve this larger context understanding what it is we're trying to do here what is what what are we building what is this purpose how do we measure success you know it can have immediate uh uh input or immediate influence on uh how a design might happen and and self-organized criticality is is one example of that so it's not just we're not just in the theoretical philosophical set here we're we're also talking nuts and bolts like okay so what might designs be like we have to think in our as we're going through the series you know we can think in our head okay that's interesting what kind of designs might reflect that concept and uh we'll we'll go into an example here with self-organized criticality so the idea there is that was coined by back back bak in 87 the term self-organized criticality and it's it's really not a controversial that that living systems and and many most systems in life complex organize in some way but the idea of self-organized criticality is that the organism itself is adjusting is keeping some kind of adjustment uh to uh to maintain a critical state and by critical state i mean a state on the ver like you can think of a saddle point so if you drop a model on us on a saddle it's going to not stay there it's going to you know it's going to change it's going to change one way or the other right so a critical state is like that that threshold where things are about to change from one way to another way and uh it turns out with you know work and information theory and other other fields of recent in recent years it turns out that uh processing uh whether it's we're talking about a computer or some other you know machine or

or a brain turns out that processing is kind of optimal in a sense when this when the system is at a this this this this critical state and some people call it on the edge of chaos because things are things can easily change and sometimes it's you can think of that threshold as a as a as a threshold of a critical state you can think of it as a threshold of the threshold we say between exploration and exploitation like should i should i go should i go find a new planet for humans to live on or should i fix the planet that you know should i fix the systems on this planet first you how do we balance exploration of the new versus using the information we have to improve what we already have so you can think of that as exploration exploitation trade-offs stability agility trade-off do we do we remain stable and use ideas from the old in the past or do we are we more agile and we're more flexible and we bring in new new ideas so it's like you can call it old new trade old new trade-off but whatever whatever trade-off you want to call it it's this sitting at the edge of going one way or the other maximally flexible of going one way or the other and it's at that threshold that level point the kind of that region of criticality that information processing seems to be maximal so if uh it's no wonder then that the human brain is is organized in such a way to be living on this threshold between agility and stability and uh now here's an example of that from like a real world example so a a system that is at a critical state is going to be maximally sensitive to input so that means that there could you know when just when that marble is sitting on the saddle just a little bump to that saddle from one little corner of its universe and right like just one little organism bumps it and maybe that marble rolls one way or the other right so that one one little input had a major impact on how the whole thing moves its trajectory into the future right but isn't that what we isn't that kind of what we have in mind for democracy i mean don't we want everyone to have access of engaging into the decision-making processes of a society and have every voice heard in at least in the sense that there's the possibility that just my voice just me doing my participation in this system might actually ripple through the system and have a uh you know a real effect a useful effect i mean i think like maybe maybe self-organized criticality can help to inform us the concept of self-organized criticality can help to inform us of what do we want from democracy or a decision-making process right you know that just makes me think about different like landslides and that's something that criticality theory and catastrophe theory has been used to study and instead of cascading failure we can think about like cascading neighborhood cleanups so a bunch of people just say today just for an hour i feel like doing a little cleanup and all of a sudden one person puts up the flag and then it's cascading locally in some just you know unspecified way but all of a sudden you're getting this this distribution

with a ton of small little meetups and then several really large sweeping changes but the total number of people cleaning up is higher because you offered the affordance and the ability for the affordance to sort of propagate that's right that's right we're talking about a propagation of a propagation of information a propagation of action and the possibility that even uh you know just one or a few individuals could start a little chain chain reaction that actually does affect in a positive way society now it's a little too it's almost too bad that sand piles were the original uh you know topic of of this of self-organized criticality because as you point out it's not really about things falling apart it's about it's about if you think of again if you think of a complex system as a system more capable of solving more challenging problems then more often you can think of self-organized criticality as a way to propagate information when it is really needed when the system needs to change uh then information is you know it ingests information from its world from its senses and can act accordingly we we just um submitted an abstract with criticality and active inference and one of the points was actually the existence of self-organized criticality implies a far from equilibrium system that's actively pumping energy in that's because it's a passive system that's not locked and loaded you don't get that kind of a non-linear response when you poke it it just stays there right right right right and i think you know maybe we won't have time to go into it right now but this also the idea of criticality is a little bit at the center of of structural organization of society through you know in the active inference theme like how is it that that that organisms and life and other systems then you know form structures will they form structures so that as you were just saying they form structures so that they can maximally cognate and those that happens to be structures that are existing in some sense near critical thresholds so and that and as also as you mentioned that of course takes energy you know it takes energy to make that happen you have to eat food to think you know right nice 14. okay so now finally we get to active inference all this discussion and we're finally getting to the point here right for his lab so um i had and i had already touched on some of this before but um it would you know today if you're going to develop a really good ai system you're and you're going to have a you have a robot saying the robot has to act in some environment it is pretty well understood that that if you program that robot to you give it a you give it a i mean traditionally you'll give it a a fitness function or some kind of valuation function and it's for example it's good if it you know you lose points if you fall through a trap door and is and you get points if you uh you know find find the piece of cake or something well that's uh that's fine for extremely simple universes that your robot might work in but as soon as you get beyond you know as soon as you get to any kind of more

realistic uh universe that your robot has to work in that pre-programming pre-programming concept just kind of falls apart it is you you it would require the the the practitioner to think ahead of all the things that the robot might encounter and then how to value certain you know value those situations in certain ways uh and that is really uh what active inference offers is a is a kind of a cognitive understanding or a mechanism by which an organism will uh uh where its fitness score is in a sense involves both uh you know achieving goals and exploring its world to for for for epistemic gain so um that's what we would like the that's how we would like to program the robot in a sense so that it can learn from it can learn on the fly from its experiences it can it can alter its actions and as it be as it becomes clear as it gathers more information from its universe as it as it meets new situations that were never never conceived of by the by the programmer that it through through an active inference or an active inference like uh you know mechanism it can learn and explore and and critically balance exploration with exploitation and then we come right back to that whole concept of criticality so you know what you would really like your robot to do is remain at that critical uh phase between exploring what's out there and making use and gold directed behavior of what's in front of it and um and uh you know that's how you could program this world this robot to act in the world and be pretty good at it you know if you if you build it well so that's what the systems of a society can help a society to do you you don't you it's worth talking about building new systems i think it would not be wise to say this checklist of like we wanted this level of education we want to want this you know to react this way in this situation react this way in this situation and this level of uh you know whatever money and this level of this and this level of that while those kinds of preferences can be a useful start society has to be alive in its moment you know in the moment as society is alive it's cognating it's it's it's it's actively uh you know comparing what it's the result of its actions to the model that is in its head and uh so active inference offers this way to uh to balance uh exploration and and uh and uh exploitation and remain critical and remain optimally cognitive right so that's part of it uh and then part of it i mean and for me this the the the idea of the embodied uh you know the three four e's uh this is what i really am attracted to in active inference is in a sense it's kind of a simple concept it's not really very complicated you know if you've studied bayesian uh theory it all it's kind of straight you know in a way it's kind of straightforward but the the you know the way fristen has connected the dots and and and and uh extended that into the bigger picture of life kind of it to me it is uh it is rich there's a there's a lot yet to be learned and gained and explored in this umbrella of active inference awesome okay so we're getting close to the end here i don't

know if maybe we should call it quits or um you know we could either go a few minutes over to just walk through these last few slides or we could um just uh do whatever you'd like to do you know i think because i can see now i'm not very good at walking fast i'm i i'm too verbose we you know it goes through a lot it goes through a lot so i think it's it's actually it's good to sometimes just take these little side paths yeah well maybe that maybe this you know leaving it at the at active inference maybe that's a good stopping point well well then maybe to kind of close on that active inference note um you know we are the active inference lab and so we always think about these sessions as a two directional highway there's people who are in the active inference community who are being exposed to some new ideas and then there's people who are being drawn in by these other things that you're bringing up which are so pressing as far as the issues as well as systems that influence everyone like governance maybe this is the first time they're hearing about active inference so it's sort of like we're uh on the cliffhanger and maybe when we come back next week we can um do a little recap and then start in where is active inference coming into play how are we going to make it specific how are we going to include people in this process what would it look like to do it's a really nice spot i think to to kind of pause okay sounds good nice so um we will hang out again uh 3 p.m on april 13th pacific time and uh that will be the second of the three sessions so thanks again for joining john this is like an awesome discussion and i hope people read the papers and we can share this first youtube link and invite people for next week's session which will be yeah at the time stated so thanks everyone see you later thanks